Unit 2: Waves and Electricity

IAS compulsory unit

Externally assessed

2.1 Unit description

Introduction	This topic covers waves and the particle nature of light and electric currents. This topic may be studied using applications that relate to electricity, for example space technology and to waves, for example medical physics. This topic also enables students to develop practical and mathematical skills.
Practical skills	In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include estimating power output of an electric motor, using a digital voltmeter to investigate the output of a potential divider and investigating current/voltage graphs for a filament bulb, thermistor and diode, determining the refractive index of solids and liquids, demonstrating progressive and stationary waves on a slinky.
Mathematical skills	Mathematical skills that could be developed in this topic include substituting numerical values into algebraic equations using appropriate units for physical quantities and applying the equation $y = mx + c$ to experimental data, using calculators to handle *sin x*, identifying uncertainties in measurements and using simple techniques to determine uncertainty when data are combined.